Project: Worked examples
Title: Voltage drop and I²R loss
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Voltage drop along a circuit in volts and per cent of the nominal voltage, the conductor power loss, and an estimate of the conductor operating temperature, for DC, single-phase and three-phase AC, from resistivity and cross-section (with temperature correction), from tabulated R and X per kilometre, or from a mV/A/m figure; parallel conductors per phase; the drop limit is a user input (the regulatory limits live in BS 7671 Appendix 4 and Regulation 525 and are not embedded).

WHAT IT CALCULATES	RELATION
Conductor resistance at its operating temperature, for n conductors in parallel	$R = \rho_T L/(A\,n)$, with $\rho_T = \rho_{20}\,[1 + \alpha\,(T_c - 20)]$
Resistance and reactance from per-length figures instead	$R = R' L/n$; $X = X' L/n$
Voltage drop on a DC circuit	$\Delta V = 2\,I\,R$
Voltage drop on a single-phase AC circuit	$\Delta V = 2\,I\,(R\cos\varphi + X\sin\varphi)$
Voltage drop on a three-phase AC circuit	$\Delta V = \sqrt{3}\,I\,(R\cos\varphi + X\sin\varphi)$
Voltage drop from a tabulated mV/A/m figure, which already carries the 2 or √3	$\Delta V = (\text{mV/A/m})\,I\,L / 1000$
Power lost in the cable	$P_{loss} = 2\,I^2\,R$ (DC and single-phase); $3\,I^2\,R$ (three-phase)
Estimated conductor temperature at this current	$T_{c,est} = T_{amb} + (T_{rated} - T_{amb})(I/I_z)^2$

Basis: Voltage drop in consumers' installations[1](525); Voltage drop (mV/A/m tables and the temperature-correction method)[1](Appendix 4, section 6); Resistance, resistivity and temperature coefficient[2](Ch. on DC and AC circuits).

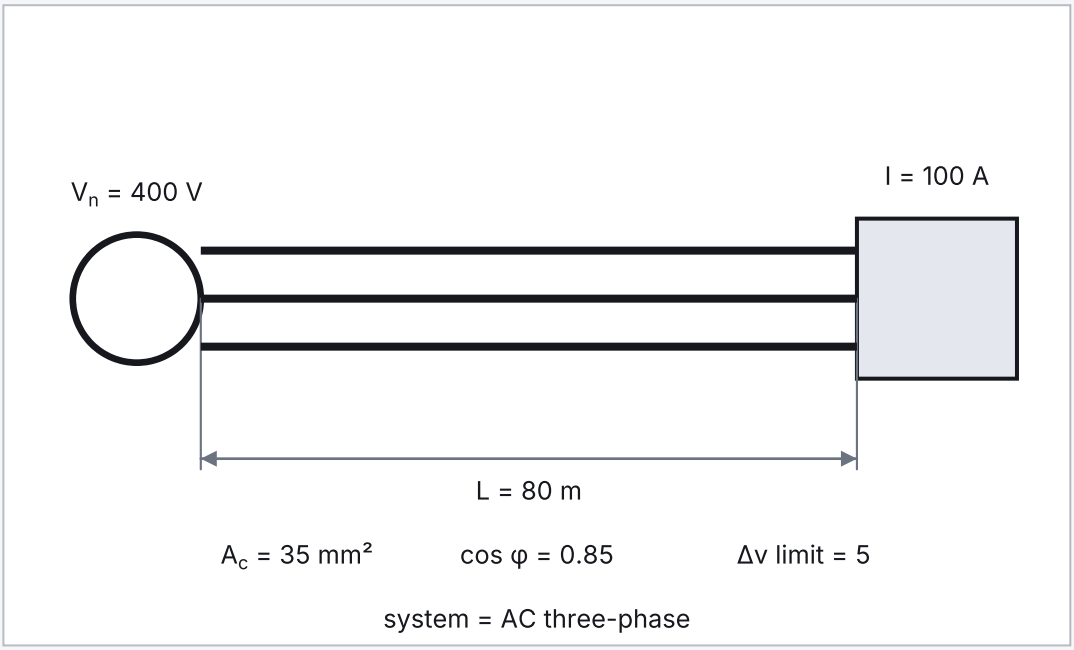


Figure 1 — A three-phase circuit: the supply, three line conductors of route length L, and the load current I. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
system	System	AC three-phase		assumption: Design convention.	any
V_n	Nominal voltage (line-to-line for three-phase)	400	V	assumption: Design convention.	12 … 1000 V
I	Design current	100	A	spec: Motor schedule MS-01: 100 A full-load current	0.1 … 5000 A
L	Route length (one way)	80	m	drawing: DRG-E-120 rev B	0.1 … 5000 m
cos_{phi}	Power factor (AC)	0.85		assumption: Design convention.	0.5 … 1
n_{par}	Parallel conductors per phase	1		assumption: Design convention.	1 … 6
data	Conductor data from	resistivity and area		assumption: Design convention.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
ρ_{20}	Resistivity at 20 °C, the effective value for the conductor resistance (default 0.0194 Ω·mm²/m: the largest class 2 maximum conductor resistance of the cable standard for 1.5 to 120 mm², so the drop is not below the cable tables; pure annealed copper is 0.01724, aluminium 0.0282 Ω·mm²/m)	0.01724	Ω·mm²/m	reference: Reference record: annealed copper 0.01724 Ω·mm²/m, 0.00393/K (Hughes)	0.01 … 0.05 Ω·mm²/m
α	Temperature coefficient of resistance	0.00393		reference: Reference record: annealed copper 0.01724 Ω·mm²/m, 0.00393/K (Hughes)	0.003 … 0.0045
A_c	Conductor cross-section	35	mm²	spec: Cable schedule CS-01: 35 mm² Cu XLPE/SWA	1 … 1000 mm²
T_c	Conductor operating temperature for the resistance	70	°C	assumption: Design convention.	20 … 90 °C
X_{km}	Reactance per km per conductor	0.08	Ω/km	spec: Cable datasheet: 0.08 Ω/km	0 … 1 Ω/km
dv_{limit}	Voltage-drop limit	5		spec: Project spec ES-3 §4: 5 % (BS 7671 Appendix 4 §6.4 default for other uses)	1 … 10
I_z	Tabulated current-carrying capacity for the installed conditions (temperature estimate)	119	A	spec: Cable schedule CS-01: I_z = 119 A for the installation method	1 … 5000 A
T_{rated}	Conductor rated temperature	70	°C	assumption: Design convention.	60 … 90 °C
T_{amb}	Ambient temperature	30	°C	assumption: Design convention.	0 … 60 °C

3. Assumptions

A1. Balanced load; conductor reactance from the stated X′ (zero when using resistivity alone unless X_{km} is given); the drop is the arithmetic (R cos φ + X sin φ) form. (Standard engineering approximation.)
A2. A mV/A/m figure is used as tabulated, with its own temperature and system convention. (Cable table convention.)
A3. The conductor temperature estimate scales the rated rise with $(I/I_z)^2$; I_z is the user's tabulated value for the installation method and corrections. (BS 7671 Appendix 4 approach, formula only.)
A4. system (System) is taken as AC three-phase. (Design convention.)
A5. V_n (Nominal voltage (line-to-line for three-phase)) is taken as 400 V. (Design convention.)

A6. $\cos_{phi}$ (Power factor (AC)) is taken as 0.85. (Design convention.)
A7. n_{par} (Parallel conductors per phase) is taken as 1. (Design convention.)
A8. data (Conductor data from) is taken as resistivity and area. (Design convention.)
A9. T_c (Conductor operating temperature for the resistance) is taken as 70 °C. (Design convention.)
A10. T_{rated} (Conductor rated temperature) is taken as 70 °C. (Design convention.)
A11. T_{amb} (Ambient temperature) is taken as 30 °C. (Design convention.)

4. Working

[9] $\sin_{phi} = \sqrt{1 - \cos_{phi}^2}$
$= \sqrt{1 - 0.85^2}$
$= 0.52678$
sin φ

[15] $\rho_T = \rho_{20} * (1 + \alpha * (T_c - 20\ degC) / (1\ K))$
$= 0.01724\ ohm·mm²/m \times (1 + 0.00393 \times (70\ degC - 20\ degC) / 1\ K)$
$= 0.020628\ ohm·mm²/m$
Resistivity at the operating temperature

[16] $R = \rho_T * L / (A_c * n_{par})$
$= 0.020628\ ohm·mm²/m \times 80\ m / (35\ mm² \times 1)$
$= 0.047149\ ohm$
Conductor resistance (one conductor, parallel paths combined)

[17] $X = X_{km} * L / n_{par}$
$= 0.08\ ohm/km \times 80\ m / 1$
$= 0.0064\ ohm$
Conductor reactance

[18] $dV = \sqrt{3} * I * (R * \cos_{phi} + X * \sin_{phi})$
$= \sqrt{3} \times 100\ A \times (0.047149\ ohm \times 0.85 + 0.0064\ ohm \times 0.52678)$
$= 7.5254\ V$
Voltage drop

[19] $dV_{pct} = 100 * dV / V_n$
$= 100 \times 7.5254\ V / 400\ V$
$= 1.8814$
Voltage drop, per cent

[20] $P_{loss} = 3 * I^2 * R$
$= 3 \times (100\ A)^2 \times 0.047149\ ohm$
$= 1414.5\ W$
Conductor power loss

[21] $u_{dv} = dV_{pct} / dv_{limit}$
$= 1.8814 / 5$
$= 0.37627$
Drop utilisation against the limit

[25] $u_{Iz} = I / I_z$
$= 100\ A / 119\ A$
$= 0.84034$
Load against the current-carrying capacity

[26] $T_{c,est} = T_{amb} + (T_{rated} - T_{amb}) * u_{Iz}^2$ [1](App. 4 §6 form)
$= 30\ degC + (70\ degC - 30\ degC) \times 0.84034^2$
$= 58.247\ degC$
Estimated conductor temperature

5. Results

SYMBOL	VALUE	DESCRIPTION
R	0.0471 Ω	Resistance
X	0.0064 Ω	Reactance

SYMBOL	VALUE	DESCRIPTION
dV	7.5 V	Voltage drop
dV_{pct}	1.8814	Voltage drop (%)
P_{loss}	1414.5 W	Power loss
$T_{c,est}$	58.2 °C	Estimated conductor temperature

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	$I \le I_z$	$u_{Iz} = 0.84$	$u_{Iz} \le 1$	0.84	[1](523)
PASS	Voltage drop ≤ limit	$u_{dv} = 0.38$	$u_{dv} \le 1$	0.38	[1](525)

7. Applicability

None.

8. Verification

Template elec.voltage-drop@1.2.0 Content hash c20fe1c4471c778d Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.